Co-operative Education Project Report

Application of artificial intelligence for tooth segmentation & numbering on 2D panoramic radiographs

(การประยุกต์ใช้ปัญญาประดิษฐ์เพื่อการแบ่งส่วนและการกำหนดหมายเลขฟันบนภาพรังสีพาโนรามาสองมิติ)

Presented by

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Academic year 2025

Cooperative Education Project Certificate

Bachelor of Engineering (Computer Engineering and Informatics)

Department of Computer Engineering Faculty of Engineering at Sriracha

Topic **Application of Artificial Intelligence for 2D tooth segmentation & numbering on 2D panoramic radiographs**
(การประยุกต์ใช้ปัญญาประดิษฐ์เพื่อการแบ่งส่วนและการกำหนดหมายเลขฟันบนภาพรังสีพาโนรามาสองมิติ)

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Has been considered and approved by

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Co-operative Education Project Report

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นาย ชัยนันท์ ชัยนนถี \ นาย ธีร์ธวัช พาอ่อน \ 2568 \ การประยุกต์ใช้ปัญญาประดิษฐ์เพื่อการแบ่งส่วนและการกำหนดหมายเลขฟันโดยอัตโนมัติบนภาพรังสีพาโนรามาสองมิติ \ The Faculty of Dentistry, Van Lang University ปริญญาวิศวกรรมศาสตร์บัณฑิต (วิศวกรรมคอมพิวเตอร์และสารสนเทศศาสตร์) \ภาควิชาวิศวกรรมคอมพิวเตอร์ \ คณะวิศวกรรมศาสตร์ศรีราชา \ อาจารย์ที่ปรึกษาสหกิจศึกษา: ไพรัช สร้อยทอง \ อาจารย์ \ 40 หน้า

บทคัดย่อ

ปัจจุบันงานทันตกรรมมีความต้องการความแม่นยำ ความรวดเร็ว และความสม่ำเสมอของผลลัพธ์มากขึ้น ส่งผลให้การนำเทคโนโลยีดิจิทัลเข้ามามีบทบาทสำคัญในการปรับปรุงกระบวนการทำงานทางคลินิกให้มีประสิทธิภาพและลดความผันผวน คลาดเคลื่อนจากการทำงานแบบแมนนวล โดยเฉพาะการประยุกต์ใช้ปัญญาประดิษฐ์ (AI) ซึ่งสามารถช่วยลดภาระงานของทันตแพทย์ผู้เชี่ยวชาญในงานที่ต้องใช้เวลาและความละเอียดสูง

งานวิจัยนี้จึงมุ่งเน้นไปที่การศึกษาการนำปัญญาประดิษฐ์ (AI) เข้ามาประยุกต์ใช้ ในงานวิเคราะห์ภาพเอกซเรย์พาโนรามาแบบสองมิติ (2D panoramic radiographs) เพื่อสนับสนุนงานด้านการแบ่งส่วนฟัน (tooth segmentation) และการระบุหมายเลขฟัน (tooth numbering) บนภาพพาโนรามา โดยได้ทำการศึกษางานวิจัยที่เกี่ยวข้องและเปรียบเทียบโมเดลเชิงลึกที่นิยมใช้ในงานลักษณะนี้จำนวน 4 กลุ่ม ได้แก่ YOLO, PANet, Transformer-based segmentation และ Mask R-CNN เพื่อประเมินจุดแข็ง–จุดอ่อนของแต่ละแนวทางและนำไปสู่การพัฒนาแนวทางของงานวิจัยให้มีความเหมาะสมกับข้อมูลจริงมากขึ้น

ผลลัพธ์ของงานวิจัย แสดงให้เห็นถึงประสิทธิภาพเปรียบเทียบของโมเดลทั้ง 4 รูปแบบ จุดเด่น ข้อดีและข้อเสีย งานวิจัยนี้จึงได้มีการพัฒนาโมเดลใหม่ขึ้นมาและมีการนำมาบูรณาการเข้ากับโปรแกรมที่พัฒนาขึ้น ผลลัพธ์ที่ได้คือแอปพลิเคชันซอฟต์แวร์ที่สามารถประมวลผลรูปภาพพาโนรามา 2มิติและแยกส่วนฟันแต่ละซี่ออกจากพื้นหลังได้โดยอัตโนมัติ อีกทั้งยังสามารถระบุหมายเลขของฟันซี่นั้นนั้นได้โดยอัตโนมัติเช่นกัน ซึ่งคาดว่าจะช่วยเพิ่มความสม่ำเสมอของผลการวิเคราะห์ ลดขั้นตอนการทำงานซ้ำ และสนับสนุนกระบวนการทำงานทันตกรรมดิจิทัลในขั้นต่อไป เช่น การวางแผนการรักษา งานทันตกรรมประดิษฐ์ และงานทันตกรรมจัดฟัน

Mr. Chaiyanan Chainontee \ Mr. Theethawat Phaorn \ 2568 \ Application of artificial intelligence for tooth segmentation & numbering on 2D panoramic radiographs \ The Faculty of Dentistry, Van Lang University \ Bachelor of Engineering (Computer Engineering and Informatics) \ Department of Computer Engineering \ Faculty of Engineering at Siracha \ Project Advisor: Pairat Sroytong \ Professor \ 40 pages

Abstract

Modern dentistry demands greater precision, speed, and consistency in results. This has led to the crucial role of digital technology in improving clinical processes, increasing efficiency, and reducing errors and inaccuracies associated with manual work. In particular, the application of artificial intelligence (AI) can significantly reduce the workload of specialist dentists in tasks requiring high precision and time.

This research focuses on studying the application of artificial intelligence (AI) in the analysis of 2D panoramic radiographs to support tooth segmentation and tooth numbering in panoramic images. It examines relevant research and compares four commonly used deep learning models for this type of task: YOLO, PANet, Transformer-based segmentation, and Mask R-CNN. The aim is to evaluate the strengths and weaknesses of each approach and develop a more suitable method for real-world data.

The research results showed a comparative analysis of four different models, highlighting their strengths and weaknesses. This research led to the development of a new model and its integration with a developed program. The result is a software application capable of processing 2D panoramic images, automatically separating individual teeth from the background, and automatically identifying each tooth by number. This is expected to improve the consistency of analysis results, reduce repetitive work, and support downstream workflow of digital dentistry, such as treatment planning, prosthetic dentistry, and orthodontics.

กิตติกรรมประกาศ

ทางคณะผู้จัดทำขอกราบขอบพระคุณต่อ ความกรุณาและการสนับสนุนจากหลายภาคส่วน รวมไปถึง ผู้มีพระคุณอย่างยิ่งของเรา The Faculty of Dentistry at Van lang University ที่ได้มอบโอกาสให้แก่ทางคณะผู้จัดทำ ได้เดินทางเข้าไปปฏิบัติงานในฐานะนักศึกษาสหกิจ พร้อมทั้งให้ความกรุณาเอื้อเฟื้อ สถานที่ เครื่องมือ อุปกรณ์ที่จำเป็นต่อพวกเรา และเหนือสิ่งอื่นใด คือ ชุดข้อมูล ภาพพาโนรามา 2 มิติ จากทางคลีนิคของมหาวิทยาลัย ซึ่งเป็นทรัพยากรที่มีความสำคัญอย่างยิ่งที่สุดต่อการดำเนินการปฏิบัติงานในครั้งนี้ ส่งผลให้โครงงานสหกิจนี้สำเร็จลุล่วงไปได้ด้วยดี

ทางคณะผู้จัดทำขอแสดงความขอบคุณต่อ Dr. Tran Hung Lam DDS, PhD, Vice Deanและ Head of Prosthodontics อีกทั้ง Dr. Viet Hoang, Researcher, Van Lang University สำหรับความกรุณาในการให้ความช่วยเหลือและสนับสนุนพวกเราอย่างเต็มที่ อีกทั้งยังต้อนรับพวกเราอย่างจริงใจและดูแลพวกเราอย่างดีในขณะช่วงที่พวกเราปฏิบัติงานอยู่ที่ Van lang University

อีกหนึ่งท่าน ผู้ที่มีบทบาทสำคัญในช่วยเหลือพวกเราให้ทำงานนี้บรรลุเป้าหมายได้อย่างลุล่วง อาจารย์ ไพรัช สร้อยทอง ประจำภาควิชาวิศวกรรมคอมพิวเตอร์ คณะวิศวกรรมศาสตร์ ศรีราชา มหาวิทยาลัยเกษตรศาสตร์ วิทยาเขตศรีราชา ที่ปรึกษาโครงงานสหกิจของพวกเราในครั้งนี้ ที่ได้ให้คำแนะนำและถ่ายทอดความรู้อันมีค่า ในการดำเนินงานแก่พวกเรา

และสุดท้ายนี้ ขอแสดงความขอบคุณต่อ คณะนักศึกษาของ The Faculty of Dentistry at Van lang University ผู้ที่คอยช่วยเหลือเราในเรื่องต่าง ๆในการดำเนินงานและการดำเนินชีวิต เราได้รับคำแนะนำที่ดีเสมอมา อีกทั้งยังต้อนรับเราอย่างอบอุ่นเหมือนบุคคลในครอบครัว คอยดูแลช่วยให้เราได้ปรับตัว และได้เรียนรู้วัฒนธรรมต่าง ๆของประเทศเวียดนามมากมายผ่านกลุ่มนักศึกษานี้ตั้งแต่วันแรกจนถึงวันสุดท้ายที่เราได้ปฏิบัติงานอยู่ที่ Van lang University

นาย ชัยนันท์ ชัยนนถี
นาย ธีร์ธวัช พาอ่อน
มีนาคม 2026

Acknowledgement

The organizing committee would like to express our sincere gratitude for the kindness and support from various sectors, including our most valued benefactor, The Faculty of Dentistry at Van Lang University, for providing us with the opportunity to work there as co-operative education students, and for graciously providing the necessary facilities, tools, and equipment. Above all, the 2D panoramic image dataset from the university clinic was the most crucial resource for this research, contributing significantly to its successful completion.

The organizing committee would like to express our sincere gratitude to Dr. Tran Hung Lam DDS, PhD, Vice Dean and Head of Prosthodontics including Dr. Viet Hoang, Researcher, Van Lang University, Van Lang University, for his kind assistance and full support, as well as his genuine welcome and excellent care during our time at Van Lang University.

Another person who played a crucial role in helping us successfully complete this project is Associate Professor Pairat Sroytong from the Department of Computer Engineering, Faculty of Engineering, Kasetsart University, Sriracha Campus, who served as our cooperative education project advisor. He provided valuable guidance and knowledge in carrying out this project.

Finally, I would like to express my gratitude to the students of The Faculty of Dentistry at Van Lang University, who assisted us in various aspects of our work and daily life. We received valuable guidance and were warmly welcomed like family. They helped us adapt and allowed us to learn so much about Vietnamese culture from the first day to the last day of our internship at Van Lang University.

Mr. Chaiyanan Chainontee
Mr. Theethawat Phaorn
March 2026

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Chapter 1

Introduction

1.1. Company Details

Van Lang University (VLU) is a higher-education institution located in Ho Chi Minh City, Vietnam. Founded in 1955. The host organization for this project is the Faculty of Dentistry, which was established in 2020, which offers instruction in the Doctor of Dental Surgery (DDS) program and focuses on developing basic dental knowledge along with clinical skills and critical thinking, as well as lifelong learning for dental education [1].

In addition, the Faculty of Dentistry at VLU has a teaching approach that supports practical teaching and research that meets real-world needs, focusing on enabling students to apply their knowledge and skills to achieve quality oral health outcomes for the community [1].

1.2. Location

The project activities and primary coordination were conducted with the Faculty of Dentistry, Van Lang University, located at the university's Main Campus: 69/68 Dang Thuy Tram Street, Ward 13, Binh Thanh District, Ho Chi Minh City, Vietnam. The Faculty of Dentistry office is listed as Room 12.10, Building G.

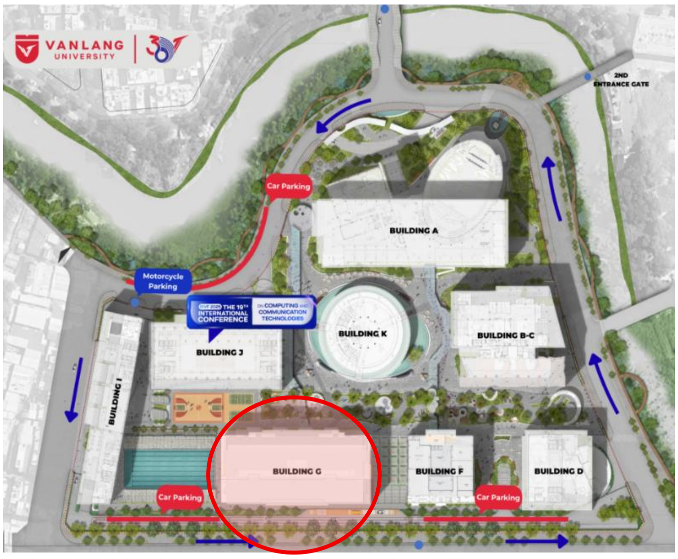


Figure 1 Map of the Vanlang University

1.3. Laboratory Members

This project was carried out with support from personnel associated with the Faculty of Dentistry at VLU. The host team structure includes:

1.3.1. Faculty leadership and academic staff (e.g., the Dean), providing academic direction and oversight .

1.3.2. Dental lecturers and staff across relevant sub-disciplines supporting training and practical activities .

1.3.3. Dental students who assisted in project-related tasks, such as supporting coordination and providing clinical feedback when assigned.

1.4. Supervisor Information

Dr. Tran Hung Lam DDS, PhD, Vice Dean and Head of Prosthodontics at Van Lang University

Email : drtranhunglam@gmail.com

1.5. Objective and Scopes of the project

1.5.1. Panoramic Radiograph

Panoramic radiography (also referred to as an orthopantomogram/OPG) is an extra-oral, two-dimensional dental X-ray technique that captures the entire dentition and the maxillomandibular complex in a single image, typically including the upper and lower jaws and surrounding structures such as the temporomandibular joints and parts of the maxillary sinuses [2]. Clinically, panoramic images are frequently acquired as an overview examination to support diagnosis and treatment planning—commonly for assessing overall dentition status, unerupted/impacted teeth (e.g., third molars), orthodontic screening, and pre-operative evaluation in oral surgery and implant planning [3]. The widespread use of panoramic radiographs is driven by their broad anatomical coverage in a single, quick acquisition and practical workflow value as a “global” view; however, the modality also exhibits well-known limitations such as superimposition, unequal magnification/distortion, and sensitivity to patient positioning, which can reduce local detail and complicate tooth boundary interpretation [4].

Figure 2 Panoramic radiograph

1.5.2. Statement of the problem

Two-dimensional panoramic radiographs are widely used in dentistry for screening and treatment planning because they provide a broad overview of the dentition and related structures in a single image [2], [4]. However, image analysis, particularly tooth segmentation and numbering, still requires expertise and is time-consuming because panoramic imaging presents unique challenges, including anatomical superimposition, variations in contrast and sharpness, patient positioning errors, and real-world clinical variability such as missing teeth or restorative materials [3]. As a result, manual work is prone to errors and can exhibit inter-operator variability among dentists, leading to inconsistent outcomes, as frequently discussed in reviews and panoramic AI studies [5-7].

Accordingly, this project explores AI-based methods for automate panoramic tooth segmentation and numbering. We study and benchmark four commonly used deep learning model families consisting of YOLO, PANet, Transformer-based segmentation, and Mask R-CNN [8-12] to understand the strengths and weaknesses based on real data and to develop appropriate approaches for clinical application.

1.5.3. Project Objectives

1). To study and benchmark four deep learning model families (YOLO, PANet, Transformer-based segmentation, and Mask R-CNN) for tooth segmentation and tooth numbering on 2D panoramic radiographs under consistent experimental settings and same data set.

2). To analyze strengths, weaknesses, and real-world failure modes (e.g., overlaps, unclear boundaries, and missing teeth) in order to guide a more practical approach for clinical panoramic data.

3). To develop and integrate an improved approach, implemented as a new AI model, into a usable program that can automatically generate segmentation and numbering outputs to support downstream digital dentistry workflows.

1.5.4. Scope of Project

1). Input data scope

2D panoramic radiographs from adult patient cases, including both complete dentitions and cases with missing teeth, reflecting real clinical variability.

2). Task scope

Tooth segmentation to separate individual teeth from the background/other structures, based on established panoramic segmentation approaches.

Tooth numbering to assign tooth numbers to segmented teeth while considering practical constraints such as missing teeth and anatomically consistent ordering.

3). Model scope

Benchmarking of four model families (YOLO, PANet, Transformer-based segmentation, and Mask R-CNN) following related comparative/benchmark studies and widely used instance segmentation foundations. Consideration of complementary directions such as multi-task learning and cross-modality transfer where appropriate.

4). Evaluation Scope

Quantitative evaluation is reported using Precision, Recall, and F1-score, computed from the confusion matrix (e.g., TP, FP, FN) to reflect prediction correctness and target coverage [13, 14].

For instance segmentation, we report mean Average Precision at IoU = 0.50 and 0.75 (commonly denoted as AP50 and AP75) and summarize these results as mAP50–75, following widely used evaluation practice in object detection/instance segmentation [13].

5). Program Scope

Development of a web application that supports panoramic image input and outputs automatic segmentation and numbering results to improve consistency and reduce repetitive manual work [6, 15].

1.6. Cooperative Education Format

Format : Team-based Project (2 members)

1.7. Cooperative Training Period

Duration : December 1, 2025, to March 27, 2026.

1.8. Cooperative Education Supervisor Information

1st (Onsite) : with Mr. Pairat Sroytong on December 1, 2025 at 1pm (GMT+7)

2nd (Online) : with Mr. Pairat Sroytong on March 25, 2026 at 7pm (GMT+7)

1.9. Working plan

Table 1 Working Plan

No.	Activities		Start Date	Finish Date	2025				2026											
					DEC				JAN				FEB				MAR			
					Week	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week
1	Literature Review		2/12/2025	20/12/2025	■	■	■													
2	Private Dataset Preparation	Acquisition	5/1/2026	6/2/2026	■	■	■	■	■	■	■	■	■							
		Annotation	6/1/2026	16/2/2026					■	■	■	■	■	■						
3	Model Benchmarking	PANet based Model [12]	18/2/2026	21/2/2026											■					
		YOLO-based Model [15]	21/2/2026	24/2/2026											■	■				
		Masks R-CNN-based Model [10]	18/2/2026	21/2/2026											■					
		Transformer based Model [9]	22/2/2026	25/2/2026											■	■				
4	Developing New Model [12]		26/2/2026	4/3/2026												■	■			
5	Model Evaluation		6/3/2026	10/3/2026													■	■		
6	Web Application Development		9/3/2026	19/3/2026													■	■	■	
7	Report and Presentation		2/3/2026	27/3/2026													■	■	■	■

Chapter 2

Literature Review

2.1. Deep learning for panoramic dental radiograph analysis

Panoramic radiographs are widely used in routine dental diagnosis and treatment planning. but automatic tooth-level analysis is challenging because of projection artifacts, intensity nonuniformity and potential occlusions between adjacent teeth. At present, AI (or deep learning) models are applied to managing panoramic X-ray images such as segmentation, identification/numbering, and much wider classification of tooth-condition. Systematic review and meta-analyses show that DL models can achieve high performance for tooth detection/segmentation but the performances can vary significantly among datasets, annotations protocols, and evaluation metrics [5], underlining the necessity of common benchmarks with reproducible experimental settings. Moreover, many studies based on panoramic radiographs also indicate that generalization is not possible to be well ensured when models are built using single-center data and tested under the same acquisition condition, encouraging more robust training strategies and cross-center validation [6, 7].

2.2. Tooth segmentation in panoramic radiographs

2.2.1. From semantic segmentation to instance segmentation

Tooth segmentation can be divided into two types: semantic segmentation (tooth and background) and instance segmentation (separate each tooth into an individual mask). Semantic segmentation can provide a rough segmentation of teeth. However, dental applications involving teeth, such as numbering and treatment planning, require high-precision segmentation with clear separation. Instance segmentation can perform this function better than semantic segmentation. A practical line of work combines encoder–decoder networks (e.g., U-Net variants) with post-processing procedures to split adjacent teeth.

For example, U-net based segmentation with morphological processing has been reported to achieve high overlap scores and significantly reduce tooth detection errors. This demonstrates that classical architecture remains effective when combined with efficient and appropriate processes [16].

2.2.2. Robust segmentation and cross-center settings

Beyond accuracy, model robustness across different acquisition devices and clinical centers remains a challenge. Two-step strategies have been proposed to improve cross-center instance segmentation: the first stage identifies approximate tooth regions, and the second stage refines high-resolution masks, reducing sensitivity to domain shifts and local artifacts [17].

2.2.3.Transformer-based segmentation for panoramic dentistry

More recently, transformer-based segmentation has been studied to better capture global context in medical images. For example, TransUNet, Swin-Unet and HiFormer integrate interest mechanisms to create remote dependency models, which may help resolve blurred boundaries and overlapping structures.[18-20]. In dental panoramic segmentation, transformer-based models have been adapted to tooth segmentation tasks, including approaches that combine semantic and instance segmentation cues through panoptic-style formulations [9]. These methods suggest that global context can complement local texture features, which is particularly beneficial in panoramic radiographs where spatial arrangements follow anatomical priors but local appearances can be degraded.

2.3. Tooth identification and FDI numbering

Tooth numbering on panoramic radiographs generally involves assigning an FDI label to each tooth. A common approach is to create a pipeline that begins by segmenting a region-of-interest (ROI), then detects tooth candidates, and finally classifies each detected tooth into an FDI category. A multi-stage pipeline using ROI segmentation, object detection, and tooth classification has been reported to achieve reliable numbering performance [21]. Similarly, Faster R-CNN-based systems have been used for automatic detection and numbering, demonstrating the possibility of predicting FDI labels directly from detected tooth regions [22].

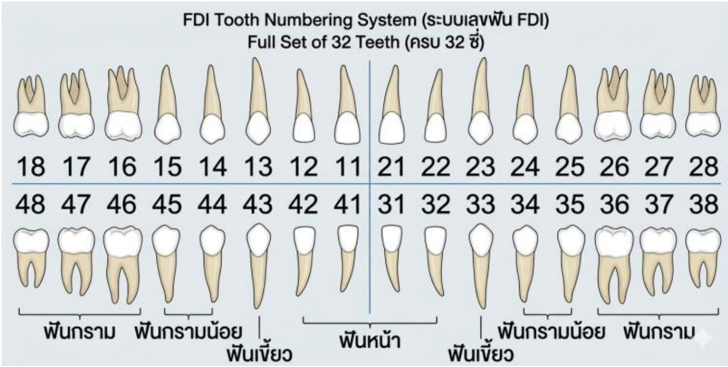


Figure 3 The FDI tooth numbering System

Pediatric scenarios (deciduous or mixed dentition) introduce additional complexity due to different tooth sets and eruption patterns, and specialized numbering systems have been proposed for pediatric panoramic radiographs [8, 23].

The major problems with tooth numbering are mostly encountered with duplicate labels, inconsistent tooth ordering, and overlapping detections. To address these problems, some studies combine DL-based detection/segmentation with explicit heuristic post-processing steps, resolve duplicates, and handle missing teeth more reliably [24]. This “deep model + rule-based correction” is highly relevant for clinical applications because it improves prediction capabilities and allows the system to incorporate domain knowledge (e.g., left-to-right ordering and arch consistency).

2.4. End-to-end frameworks for joint segmentation and numbering

The combination of instance segmentation and numbering offers a new and improved approach, allowing a single framework to generate both mask and tooth identities. A representative benchmark study compared multiple state-of-the-art instance segmentation architectures (including Mask R-CNN and PANet-style variants) for tooth instance segmentation and numbering on panoramic images, reporting that PANet-based configurations achieved strong performance for both segmentation and numbering under the same experimental protocol [12]. Other studies investigate encoder–decoder models augmented with attention mechanisms and pretrained backbones, aiming to improve segmentation/identification under overlapping teeth by integrating multi-scale features and attention gating [25]. In addition, transfer learning across modalities has been studied. For example, systems pretrained using CBCT-derived panoramic representations have been proposed to improve generalization when real annotated panoramic data are limited [26]. Finally, multi-task extensions incorporate additional results, such as tooth status assessments, demonstrating a trend toward a more integrated panorama analysis pipeline than segmentation and numbering [15].

2.5. Tools and Techniques in Prior work

Prior studies on tooth segmentation and tooth numbering in 2D panoramic radiographs typically follow a similar practical workflow:

1). Creating instanced or multi-class ground truth data.

2). applying preprocessing/augmentation to mitigate panoramic variability

3). training models using standard deep learning toolchains

4). post-processing and evaluation with established metrics

However, some studies do not explicitly report the full toolchain or the specific tools used [5-7].

2.5.1. Annotation tools and labeling protocol

Segmentation/instance segmentation on 2D panoramic images is commonly requires the use of mask labels (e.g., polygon/instance masks or multi-class masks) Several studies have indicated that this process is performed manually by dental experts or researchers trained by dentists, sometimes accompanied by review steps to improve label consistency [5, 6].

2.5.2. Preprocessing and augmentation

Panoramic images suffer from modality-specific issues (e.g., contrast/sharpness variation, positioning-related distortion, and artefacts) preprocessing is often used to standardize inputs (e.g., scaling/intensity/contrast adjustment), and augmentation is applied to improve robustness under real-world variability [3, 6]. This approach is often preferred for dealing with difficult cases (e.g., overlapping teeth or indistinct borders) [9, 17].

2.5.3. Training toolchain

In terms of model development, many works adopt widely used model families for segmentation/detection, including YOLO, PANet, Transformer-based segmentation, and Mask R-CNN, as also reflected in comparative/benchmark studies [8-12]. However, some studies use different approaches, such as pretrained backbone or knowledge transfer, to optimize performance under real data constraints (e.g., limited label data or domain gap data) [26].

2.5.4.Post-processing & evaluation practice

For tooth numbering, Previous research has often incorporated rule-based or heuristic constraints or multi-stage pipelines to address duplicated labels, anatomically inconsistent ordering, and missing-teeth scenarios [21, 22, 24], There is also a trend toward multi-task designs that extend beyond segmentation/numbering to include additional clinically relevant outputs [15].

Evaluation commonly reports Precision, Recall, and F1-score, and uses AP at different IoU thresholds such as AP50 and AP75, which can be summarized as “mAP50–75” following standard evaluation definitions [13, 14].

2.6. Summary and research gap

Despite ongoing research and progress, the literature reveals persistent gaps for real-world panoramic tooth segmentation and numbering. Firstly, most research still uses small datasets, primarily consisting of complete sets of 32 teeth, which can lead to difficulties in addressing cases of missing teeth and abnormalities, which are common in real-world clinics. Second, the vastly different evaluation protocols across various research studies (e.g., Dice/IoU vs. COCO mAP, data aggregation per tooth vs. data aggregation per image) make direct comparisons difficult. Third, while end-to-end models are attractive, heuristic corrections and anatomical priors remain important for reducing clinically unacceptable numbering errors.

These three observations inspired us to design our research as follows:

1). adopting a standardized instance segmentation + numbering formulation consistent with COCO-style annotations,

2). evaluating multiple model families under controlled settings (e.g., PANet/Mask R-CNN vs. one-stage alternatives), and

3). reporting both segmentation and numbering metrics, including error analyses that reflect clinically relevant failure modes.

Chapter 3

Material and Method

3.1. Research Overview

This study focuses on developing an AI-based program for analyzing 2D panoramic radiographs by performing tooth segmentation and tooth numbering on the images. The AI model is trained to learn from real-world clinical data collected from a private dental clinic, obtained through collaboration with Van Lang University. The proposed system aims to improve result consistency, reduce variability from manual analysis, and support downstream digital dentistry workflows such as treatment planning, prosthodontic design, and orthodontic applications.

3.2. Methods

3.2.1. Dataset acquisition and annotation

1,247 2D panoramic radiographs were used in this study. The dataset consists of adult patient cases and includes both complete dentitions and cases with missing teeth. The data were collected through collaboration with Van Lang University.

Figure 4 Example of complete dentition cases

Figure 5 Example of cases with missing teeths

For ground-truth annotation, two members of the research team performed the labeling after receiving training provided by the Faculty of Dentistry at Van Lang University. Annotations were created using the CVAT platform, which was deployed on a local server running on the researchers' own computer. After annotation was completed, the labeled results were reviewed and verified by dental students and dental faculty members from Van Lang University to improve annotation quality and consistency.

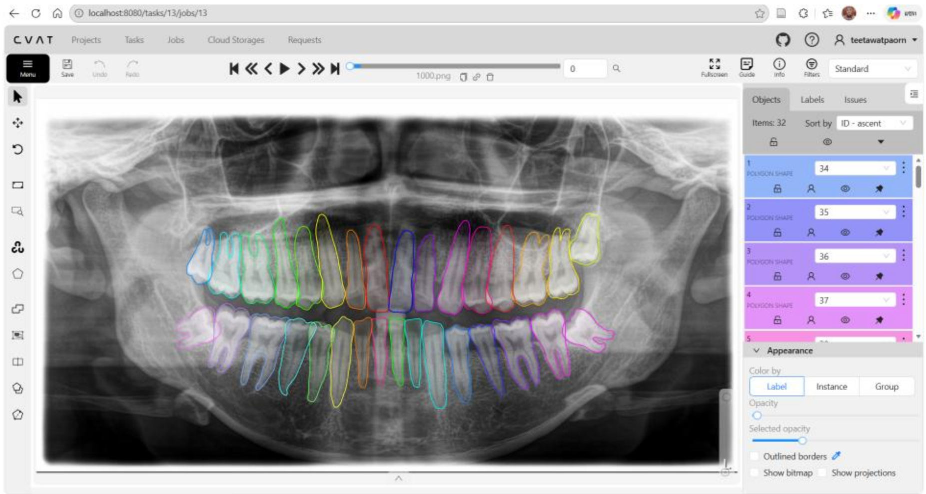


Figure 6 CVAT local annotation platform

3.2.2. Split Dataset

We divided the total dataset of 1247 into three parts for training, validation, and testing, with ratios of 70%, 15%, and 15% respectively. This dataset will be used with our four base models and a new model we developed to prevent data selection bias.

3.2.3.Base Line Model

We chose four baselines (YOLO, PANet, Mask2Former, and Mask R-CNN) to cover the main design directions used in modern instance segmentation and to keep the comparison easy to interpret. Panoramic radiographs are challenging (teeth appear at different scales, boundaries can be faint, and adjacent teeth often touch or overlap) [3, 9, 17]. Because each model family handles these issues differently, a single baseline would not be informative. [12, 17]

With these four baselines, we can:

1). compare one-stage, two-stage, and transformer-based formulations [8-10, 17].

2). see how much stronger feature aggregation (PANet) helps beyond a standard two-stage model [12].

3). test whether transformer-based segmentation improves separation in crowded tooth regions, all under the same data split and evaluation protocol [9, 17].

YOLO-based Model (One-stage Instance Segmentation)

Why we included it: YOLO is a practical, deployment-oriented one-stage family. It gives us a clear reference point for inference efficiency and shows what level of accuracy is achievable without an explicit region-proposal step. [8]

Method summary: The model predicts bounding boxes, class scores, and an instance mask for each detected tooth in a single forward pass. [8] At inference time, we apply a confidence threshold and non-maximum suppression (NMS) to remove low-confidence and duplicate detections. The remaining instances are rendered as overlays on the panoramic image and used to display the corresponding tooth numbers. [8]

1). Single-pass prediction (box, class, and mask) without a proposal stage. [8]

2). A speed/throughput baseline that reflects real deployment constraints. [8]

3). Useful for illustrating the speed-accuracy trade-off on panoramic radiographs. [8]

PANet-based Model (Two-stage Baseline with Enhanced Feature Aggregation)

Why we included it: PANet improves multi-scale feature aggregation compared with a standard FPN-based two-stage pipeline. This matters for panoramic radiographs, where inter-tooth gaps can be very small and boundaries are often weak; better feature propaga tion helps preserve fine spatial detail. [3, 12, 17]

Method summary: The pipeline follows a two-stage design: a region proposal network produces candidate regions, RoI features are extracted, and separate heads predict the class, bounding box, and instance mask. PANet extends the feature pyramid with bottom-up path aggregation so that localization detail from lower layers is better propagated to higher pyramid levels. [12]

1). Two-stage pipeline with enhanced multi-scale feature fusion (beyond a vanilla FPN). [12]

2). Often helps when instances are tightly packed and boundaries are subtle. [3, 9, 17]

3). A strong two-stage reference between Mask R-CNN and the proposed cascade model. [12]

Mask R-CNN (ResNet-50 + FPN) - Canonical Two-stage Baseline

Why we included it: Mask R-CNN is a widely used and well-understood benchmark for instance segmentation. It provides a stable reference to judge whether more advanced designs (stronger aggregation, transformers, or cascades) deliver meaningful gains on the same dataset. [10, 12]

Method summary: The model uses an RPN to generate proposals, ROI Align to extract aligned RoI features, and separate heads for classification, box regression, and mask prediction. The FPN neck supplies multi-scale feature maps, which is important in panoramic images where tooth size and appearance vary across jaw regions. [10, 12]

1). Standard two-stage design (RPN + ROI Align + mask head). [10, 12]

2). FPN supports multi-scale representation across different tooth sizes. [3, 12, 17][1]

3). A clear, interpretable benchmark for comparing newer model designs. [10, 12]

Mask2Former (Transformer-based Query Segmentation)

Why we included it: Mask2Former represents a modern transformer-based, query-driven approach to mask prediction. [9, 18-20] It is well suited to scenes with many nearby instances (such as teeth) because global context and one-to-one assignment during training can reduce duplicates and improve separation. [3, 6, 9]

Method summary: A pixel decoder produces multi-scale features, and a transformer decoder outputs a fixed set of mask queries. [9, 18-20] Each query predicts a class label and an instance mask. [9] During training, predicted queries are matched to ground-truth instances using bipartite (Hungarian) matching, encouraging a one-to-one correspondence and discouraging duplicate outputs.

1). Query-based prediction: each query outputs a class label and an instance mask. [9]

2). Hungarian matching provides one-to-one assignment during training, which helps reduce duplicates.

3). Global context is useful for resolving ambiguous boundaries in crowded tooth regions. [3, 6, 9]

3.2.4. Our Model

Rationale for Selecting the Main/Proposed Model: HTC (Swin-T + FPN)

Panoramic radiographs often have low contrast and overlapping anatomy, so tooth boundaries can be ambiguous. [3, 6] In these cases, single-pass mask prediction may miss subtle edges or merge adjacent teeth. [9, 17] We therefore use Hybrid Task Cascade (HTC) as the main model because it refines predictions over multiple stages. Each stage takes the previous stage's outputs and improves both the bounding boxes and the masks, which typically produces cleaner boundaries and more stable instance separation in crowded areas. [9, 17]

We pair HTC with a Swin Transformer-Tiny backbone to strengthen contextual modeling through shifted-window self-attention while keeping a hierarchical representation. [18-20] Together with FPN, the model retains robust multi-scale features, helping it handle both large molars and smaller anterior teeth within the same panoramic image. [3, 17]

1). Multi-stage cascade refinement tends to sharpen mask boundaries and separate adjacent teeth more reliably. [9, 17]

2). Swin-T improves contextual representation compared with conventional CNN backbones in dense regions. [9, 18, 20]

3). FPN maintains multi-scale robustness for the wide range of tooth sizes in panoramic views. [3, 17]

Proposed/Main Method Description: HTC (Swin-T + FPN)

HTC stacks multiple detection and mask heads in a cascade. At each stage, the model refines bounding boxes and masks using the outputs from the previous stage, leading to progressively improved instance segmentation. [9, 17] Swin-T captures context efficiently via shifted-window self-attention, and FPN merges features across scales. [17-20] In our experiments, HTC (Swin-T + FPN) is treated as the main model and evaluated against all baselines using the same data split, training budget, and evaluation metrics. [12, 17]

3.3. Experimental Setup

All Models were trained and evaluated on a workstation equipped with an NVIDIA GeForce RTX 4060 GPU (CUDA-enabled). To ensure a fair comparison, we used the same dataset split and evaluation code across all methods. [12] Each method was trained for 80 epochs, and we report test-set performance using the best checkpoint selected on the validation set. [12]

3.4. Evaluation Metrics (Pixel-wise)

In addition to instance-level COCO mask AP, we report pixel-wise classification metrics by merging all tooth instances into a single binary teeth mask (teeth vs. background). [13] Let TP, TN, FP, and FN denote the number of pixels that are true positives, true negatives, false positives, and false negatives, respectively: [14]

TP: pixels labeled as teeth in the ground truth and predicted as teeth.

TN: pixels labeled as background in the ground truth and predicted as background.

FP: pixels labeled as background in the ground truth but predicted as teeth.

FN: pixels labeled as teeth in the ground truth but predicted as background.

Overall Accuracy: measures overall pixel correctness**.** [14]

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

Specificity (True Negative Rate): measures correct background identification (reducing FP). [14]

$$Specificity = \frac{TN}{(TN + FP)}$$

Precision: measures the reliability of predicted tooth pixels (penalizing over-segmentation). [14]

$$Precision = \frac{TP}{(TP + FP)}$$

Recall (Sensitivity / True Positive Rate): measures the completeness of recovered tooth regions (penalizing under-segmentation). [14]

$$Recall = \frac{TP}{(TP + FN)}$$

F1-score: summarizes the precision–recall trade-off via the harmonic mean, providing a single indicator that balances both error types [14]

$$F1\ score = \frac{2 \times Precision \times Recall}{(Precision + Recall)}$$

3.5. Web Application and Deployment

3.5.1. System Architecture

The ToothInSeg platform follows a three-tier architecture with explicit trust boundaries. The frontend provides the user interface, the backend enforces authentication and orchestrates inference, and the GPU inference service performs model execution. MongoDB is used for persistent metadata and user history, while Cloudinary is used to store images and overlays via private delivery.

1). Frontend: Next.js (TypeScript) deployed on Vercel for authentication, image upload, visualization, and history browsing.

2). Backend: Java Spring Boot deployed on Railway as the system's security boundary (JWT validation, access control, and orchestration).

3). Inference service: Python service packaged as a Docker container and deployed on RunPod (GPU) for model inference.

4). MongoDB Atlas: stores user accounts and lightweight prediction history metadata (no large binary payloads).

5). Cloudinary: stores input radiographs and overlay outputs using authenticated (private) delivery; the backend generates signed URLs.

3.5.2. End-to-End Prediction Workflow

The backend receives the uploaded image, verifies authorization, forwards raw bytes to the inference service, post-processes the results into clinically interpretable summaries (present/missing FDI), and persists a lightweight history record.

1). The user logs in and receives an access token (JWT).

2). The user uploads a panoramic X-ray image (single image or batch) via the web interface.

3). The frontend sends the image to the backend endpoint POST /api/predict as multipart/form-data (file and optional path metadata).

4). The backend verifies the JWT, normalizes the optional path metadata, and forwards the raw image bytes to the inference service (/predict_raw) with an internal API key (X-API-KEY).

5). The inference service returns structured prediction results (JSON). The backend performs lightweight post-processing to derive per-tooth summaries (present/missing FDI).

6). The input images and overlay outputs has upload to Cloudinary (authenticated/private), saves metadata and history to MongoDB, and returns the final response (including signed URLs) to the frontend.

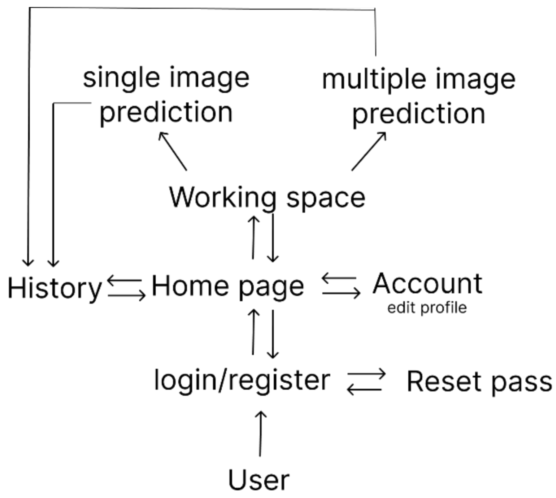


Figure 7 Workflow of Web Application

3.5.3. REST API (Frontend, Backend)

Table 2 lists the endpoints used by the frontend for user authentication, prediction, and history retrieval.

Method	**Endpoint**	**Purpose**	**Auth**
POST	/api/auth/register	Create a new user account.	Public
POST	/api/auth/login	Authenticate and return accessToken (JWT).	Public
GET	/api/me	Fetch the current user profile.	JWT
PUT	/api/me	Update the current user profile (e.g., fullName).	JWT
POST	/api/predict	Upload an image and run inference (file, optional path).	JWT

Method	Endpoint	Purpose	Auth
GET	/api/history	List prediction history for the authenticated user.	JWT
GET	/api/history/[15]	Fetch a single history record (scoped to userId).	JWT

In addition to the endpoints above, the system supports password recovery (forgot/reset password). These endpoints are part of the production-ready user management but do not affect the model inference pipeline.

3.5.4. Privacy and Security

Panoramic radiographs can be sensitive. The application therefore adopts privacy-by-design measures to limit exposure of patient images and to prevent unauthorized access to the inference service.

1). Backend as the security boundary: the frontend never calls the inference service directly.

2). JWT-based stateless authentication on the backend; non-public endpoints require authorization.

3). Server-to-server protection for inference: the backend forwards image bytes to the ML endpoint using X-API-KEY, which is stored only on the server.

4). Private media delivery: Cloudinary assets are stored using authenticated (private) delivery; the backend generates signed URLs for retrieval.

5). Data minimization: MongoDB stores only lightweight metadata and Cloudinary references, not large binary payloads.

6). Input path normalization: optional path metadata is sanitized before being stored to prevent traversal or unintended disclosure.

3.5.5. Deployment Summary

1). Vercel hosts the Next.js frontend (configured with NEXT_PUBLIC_API_BASE to point to the backend).

2). Railway hosts the Spring Boot backend (configured with MongoDB, JWT, ML service URL/API key, and Cloudinary settings).

3). RunPod hosts the Dockerized inference service (GPU) exposing /predict_raw.

4). MongoDB Atlas stores user and history documents; Cloudinary stores private images and overlays.

3.5.6 Appendix: Key Source Files for Reproducibility

1). Frontend: src/lib/api.ts — centralized REST client and batch prediction concurrency.

2). Backend: controller/PredictController.java — predict + history endpoints, post-processing, Cloudinary integration.

3). Backend: client/MlClient.java — server-to-server inference call with X-API-KEY.

4). Backend: service/CloudinaryService.java — authenticated upload and signed URL generation.

5). Backend: config/SecurityConfig.java — JWT stateless security configuration and public endpoints.

Chapter 4

Results

4.1 Quantitative Results

This section reports the quantitative performance of all five models on the held-out test set of 187 images (15% of the full dataset) after 80 training epochs. The comparison is summarized in Table 2.

Among the evaluated methods, HTC (Swin-T + FPN) achieved the highest F1-score (94.96) and the highest mAvP (80.20), indicating the best overall balance between accurate mask delineation and instance-level segmentation performance on the same test set.

Table 3 Quantitative performance of all five models on the same data set.

Method (best ckpt)	Acc.	Spec.	Prec.	Rec.	F1	mAvP	AP50	AP75
Mask R-CNN (ResNet-50 + FPN) best@38	98.67	99.21	94.37	94.77	94.57	78.00	98.30	95.30
Mask2Former (ResNet-50) best@10	98.72	99.24	94.58	94.99	94.78	80.10	98.40	95.90
YOLOv8	98.47	98.76	91.69	96.44	94.01	73.24	97.97	92.24
PANet (Mask R-CNN)	98.63	98.98	92.93	96.12	94.50	76.87	97.07	94.24
HTC (Swin-T + FPN) best@17	98.76	99.24	94.61	95.32	94.96	80.20	98.30	96.10

4.2 Qualitative Results

To preserve visual detail, the qualitative comparison is presented as three separate figures, with one representative case shown per figure.

4.2.1 Representative Case 1: Complete Dentition

In the case of complete tooth detection, all models were able to detect all teeth, but HTC (Swin-T + FPN), Mask2Former, and Mask R - CNN (RestNet50 + FPN) yielded mask boundary results closest to Ground Truth. While YOLOv8, despite high recall coverage, sh owed slight discrepancies in tooth edge sharpness compared to the multi-step improvement in the HTC model.

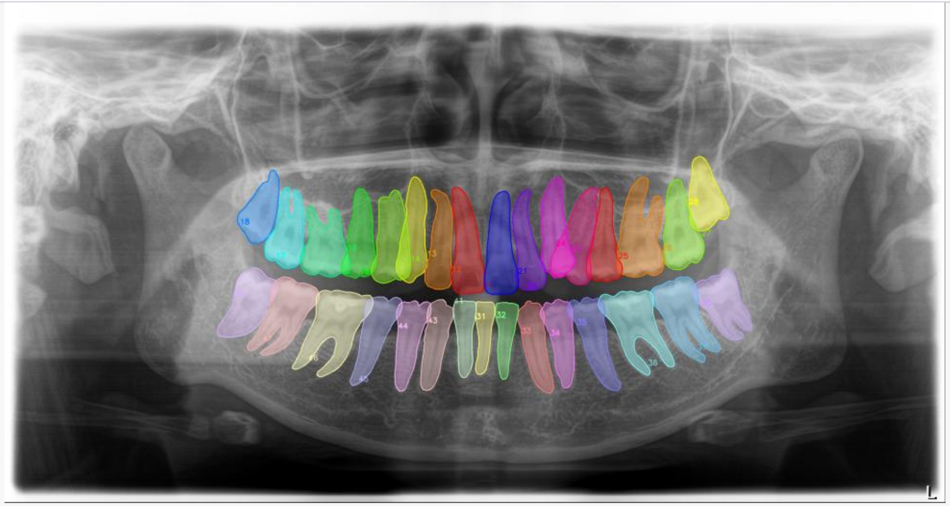


Figure 8 Ground Truth (Complete Dentition)

Figure 9 Mask R-CNN (RestNet50 + FPN)[10](Complete Dentition)

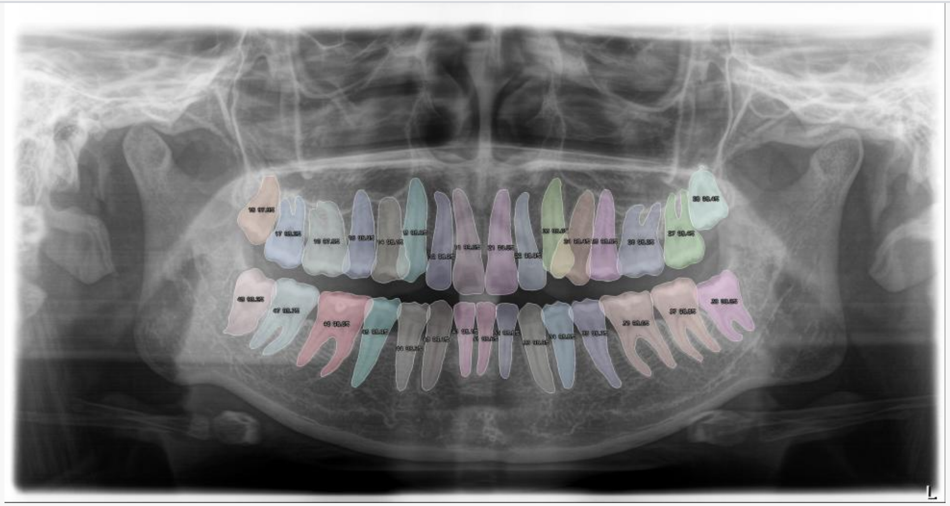


Figure 10 Mask2Former [9](Complete Dentition)

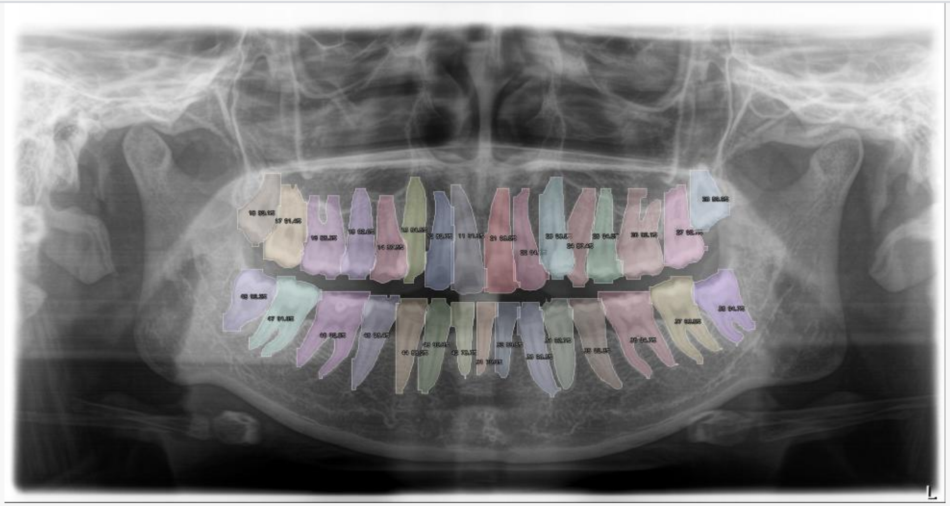


Figure 11 YOLOv8 [15] (Complete Dentition)

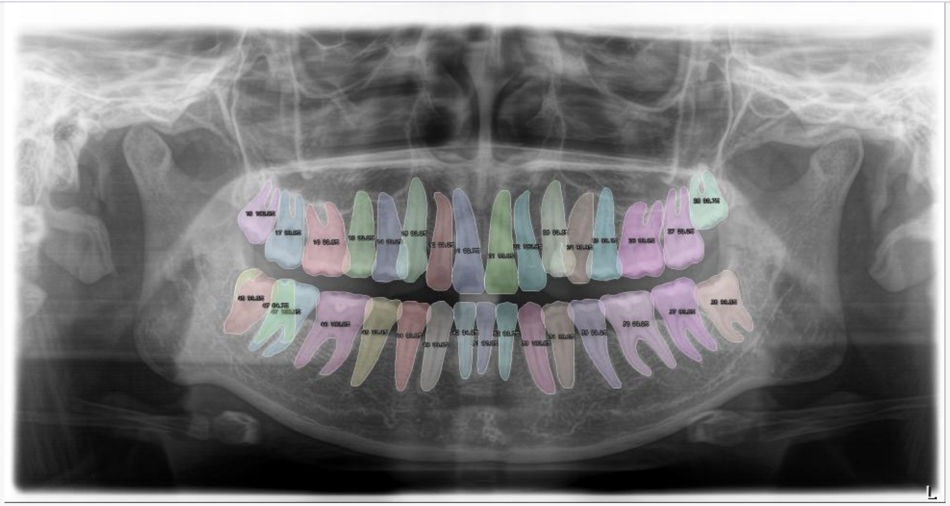


Figure 12 PANet [12] (Complete Dentition)

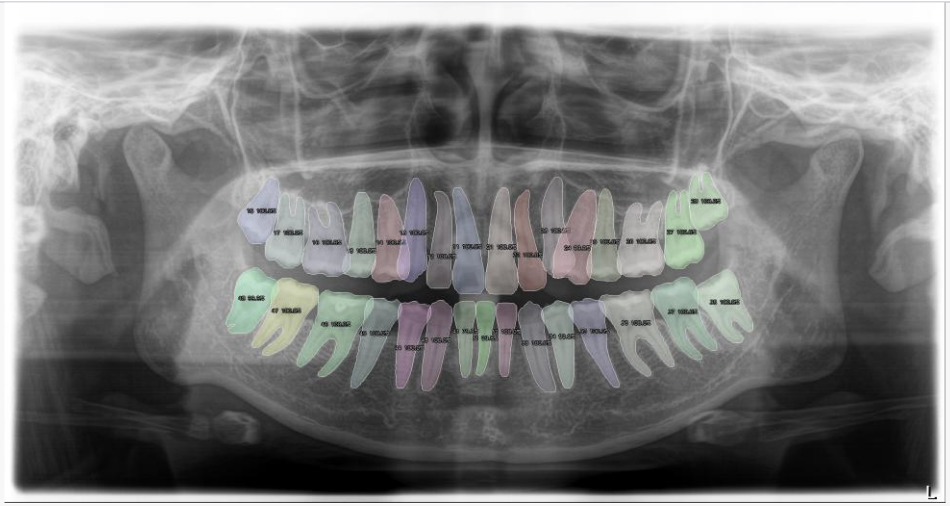


Figure 13 HTC (Swin-T + FPN) [12, 18] (Complete Dentition)

4.2.2 Representative Case 2: Missing Teeth

Case 2 illustrates the robustness of almost all models in handling gaps from tooth loss. The models can accurately identify m issing tooth areas without generating false positives in the gap. Conversely, single-step models like YOLOv8 often struggle in low-contrast areas of the gap, sometimes leading to the detection of mask fragments that don't accurately represent the actual tooth.

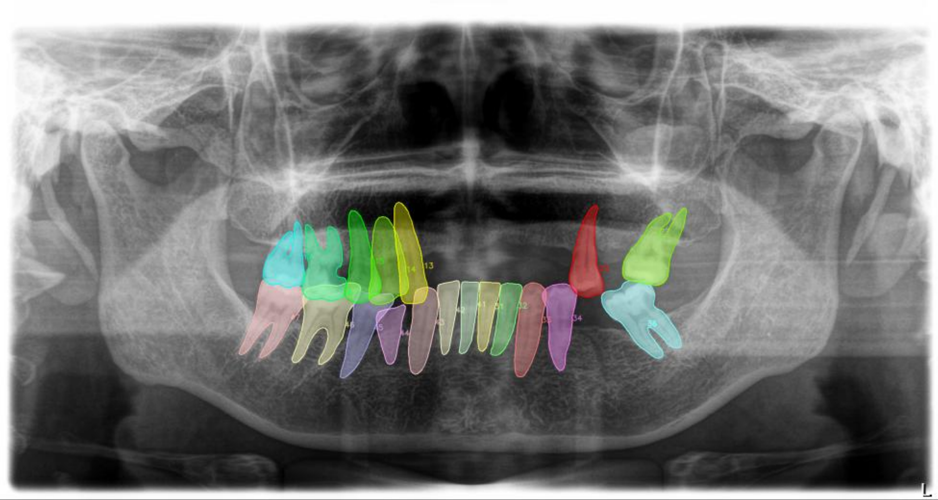


Figure 14 Ground Truth (Missing Teeth)

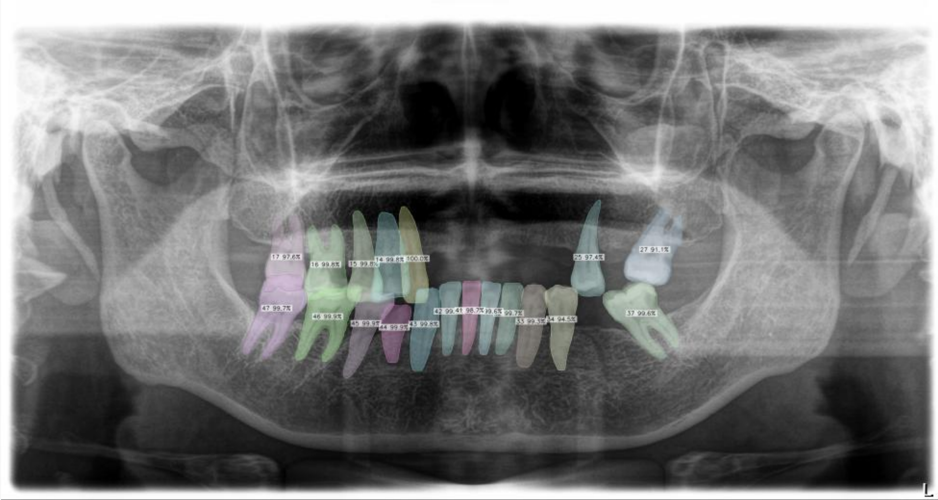


Figure 15 Mask R-CNN (RestNet50 + FPN) [10] (Missing Teeth)

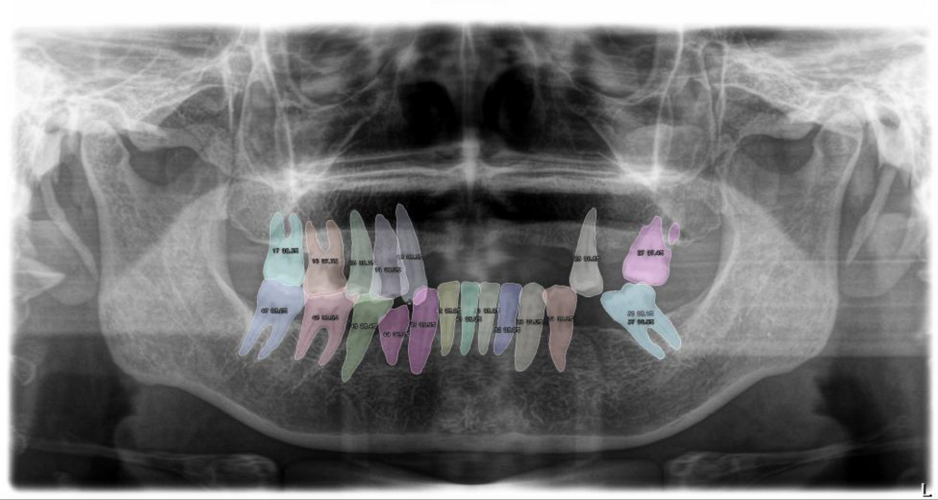


Figure 16 Mask2Former [9] (Missing Teeth)

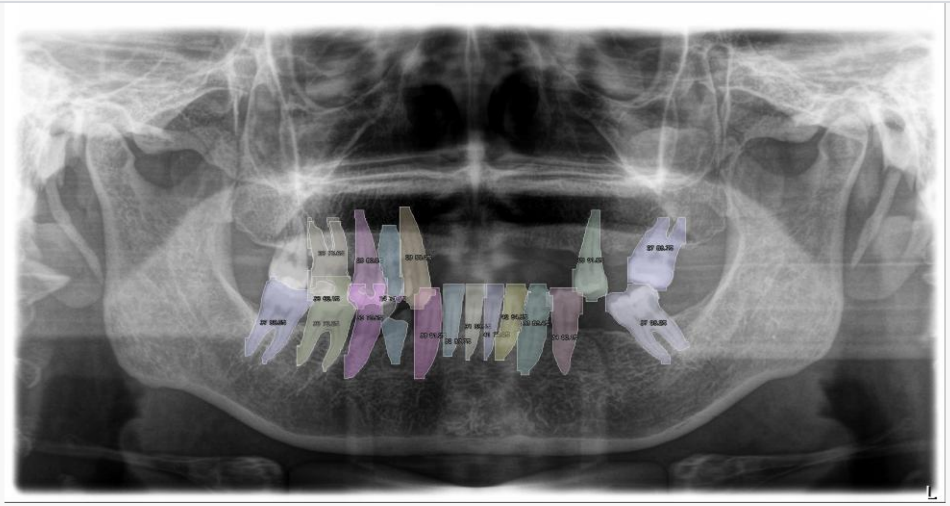


Figure 17 YOLOv8 [15] (Missing Teeth)

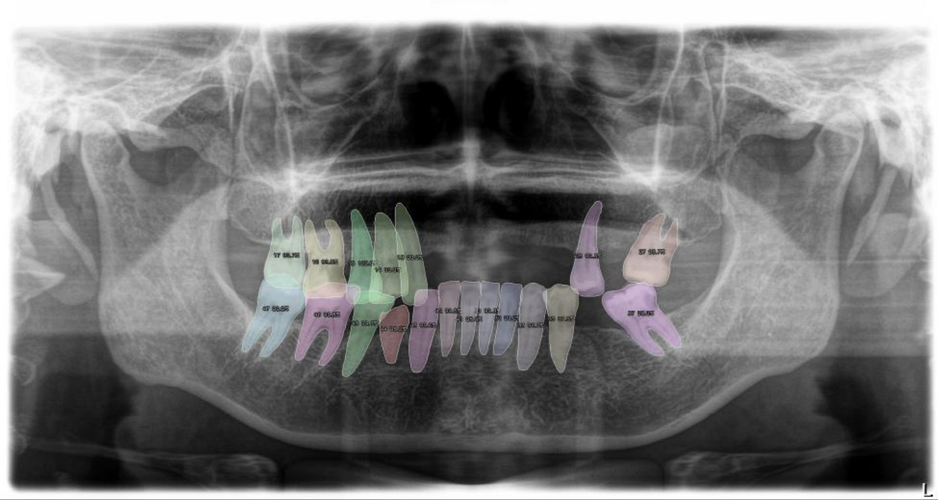


Figure 18 PANet [12] (Missing Teeth)

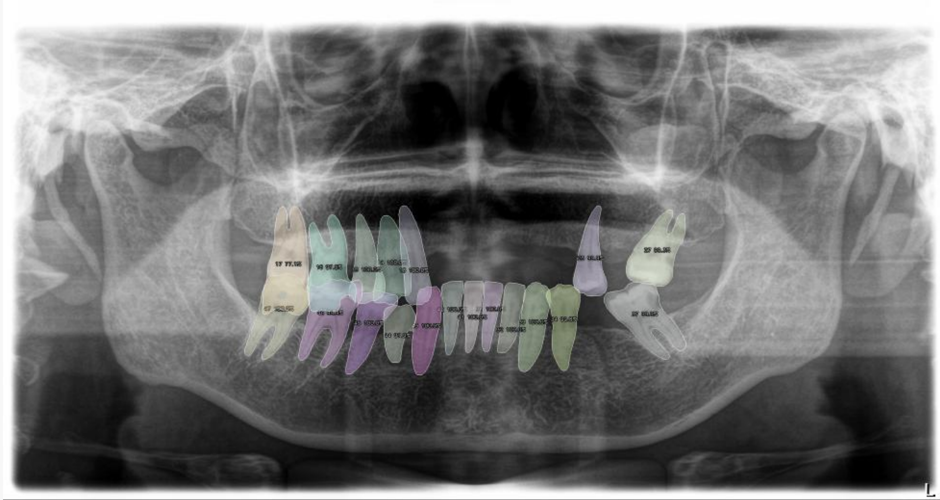


Figure 19 HTC (Swin-T + FPN) [12, 18] (Missing Teeth)

4.2.3 Representative Case 3: Crowded or Overlapping Teeth

Case 3 highlights the advantage of the Swin Transformer backbone in the HTC model for resolving crowded regions. The hierarchical contextual representation allows for superior separation of adjacent teeth that appear partially merged in the radiograph. Standard CNN-based baselines, such as Mask R-CNN, tend to produce overlapping mask boundaries or merged instances in these high-density areas, whereas HTC maintains distinct and accurate instance masks.

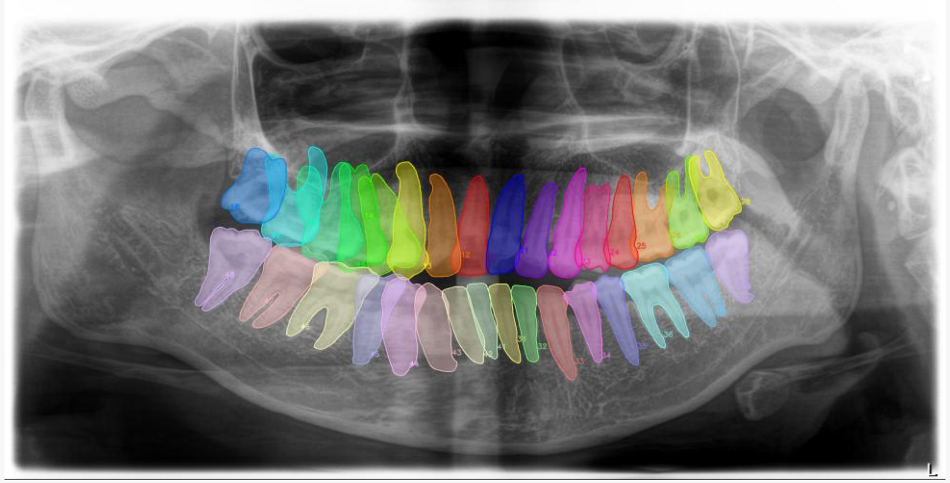


Figure 20 Ground Truth (Crowded or Overlapping Teeth)

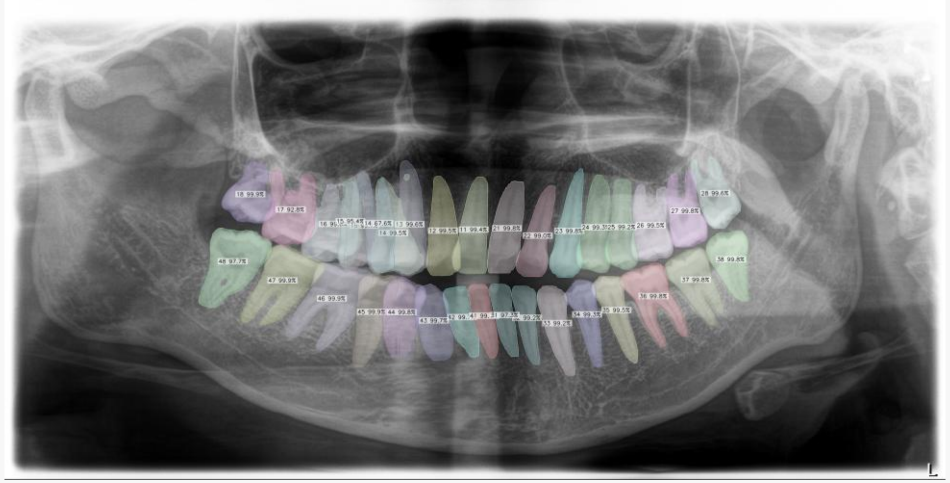


Figure 21 Mask R-CNN (RestNet50 + FPN) [10] (Crowded or Overlapping Teeth)

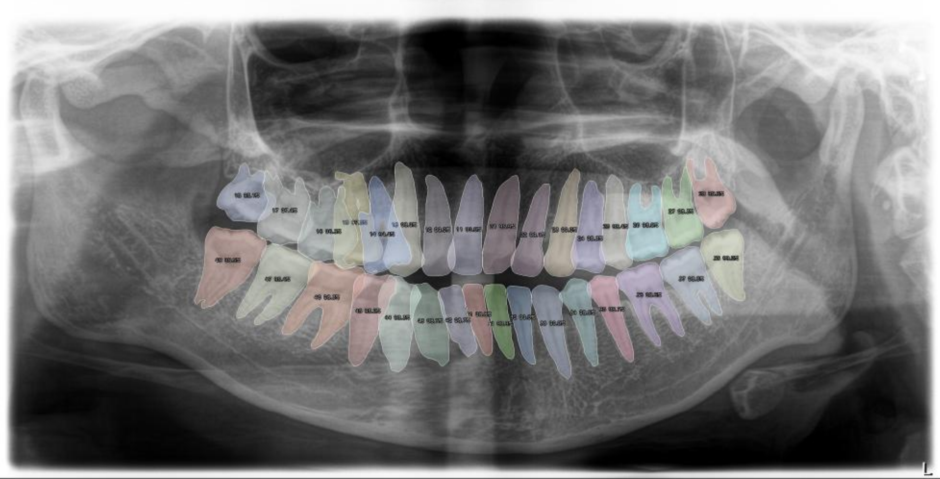


Figure 22 Mask2Former [9] (Crowded or Overlapping Teeth)

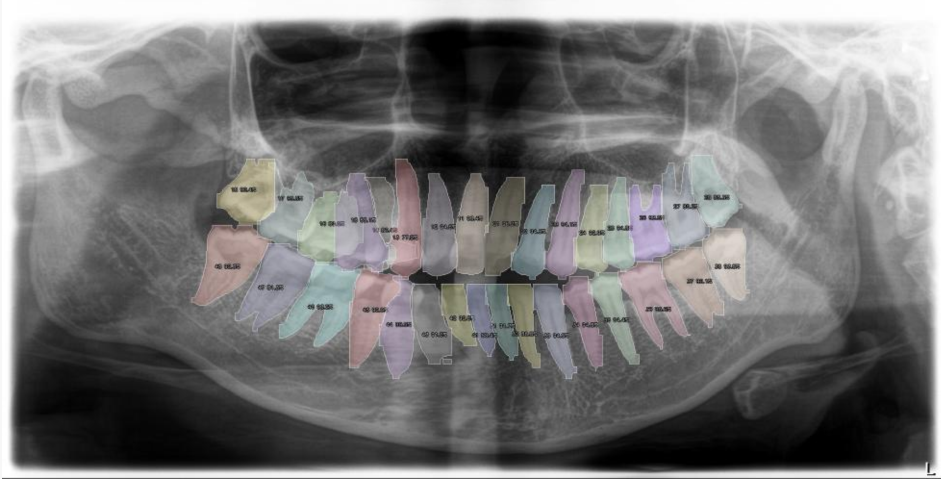


Figure 23 YOLOv8 [15] (Crowded or Overlapping Teeth)

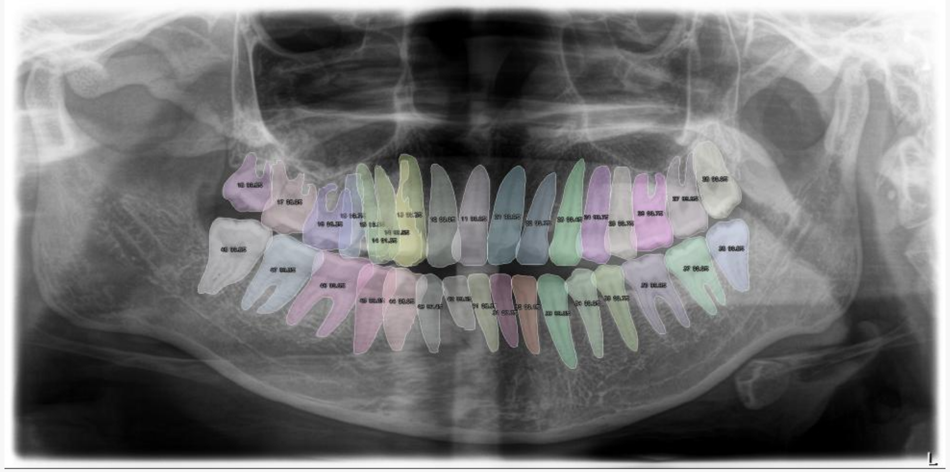


Figure 24 PANet [12] (Crowded or Overlapping Teeth)

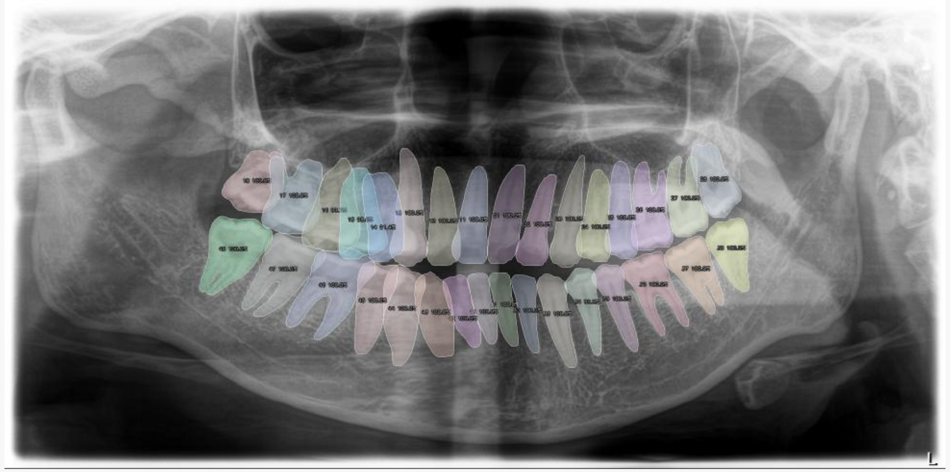


Figure 25 HTC (Swin-T + FPN) [12, 18] (Crowded or Overlapping Teeth)

4.3 Post-processing of the Best-Performing Model

Among the five evaluated models, HTC (Swin-T + FPN) achieved the strongest overall performance and was therefore selected for an additional post-processing stage. Although the raw predictions were already strong, some challenging panoramic radiographs still contained two practical error types: repeated predictions assigned to the same tooth region and very small fragment-like masks that did not correspond to a valid tooth structure.

The post-processing pipeline was designed to suppress these errors while preserving anatomically plausible teeth. First, unstable detections were filtered using mask size, connected-component cleaning, bounding-box height, and fill ratio so that tiny residual pieces around tooth boundaries were removed. Next, highly overlapping masks were compared using overlap ratio, IoU, centroid distance, and confidence score. When two masks described the same physical tooth, the algorithm retained the more reliable instance and discarded the duplicate one. Finally, local spatial consistency along the dental arch was checked so that the remaining predictions followed a more realistic tooth arrangement. Figures 11 and 12 present representative before-and-after examples of duplicate removal and fragment suppression.

4.3.1 Removal of duplicate tooth predictions

In some cases, the raw HTC output produced two highly similar masks over the same tooth or over nearly identical regions. To resolve this issue, the post-processing step compared overlapping instances and removed the one with lower reliability, based on confidence, mask overlap, centroid proximity, and compatibility with neighboring teeth. This reduced repeated labels and produced a cleaner one-to-one correspondence between each tooth region and its final prediction.

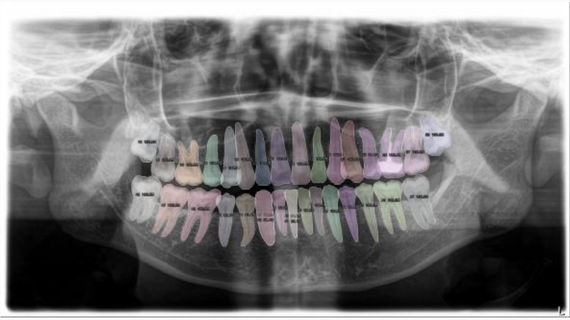
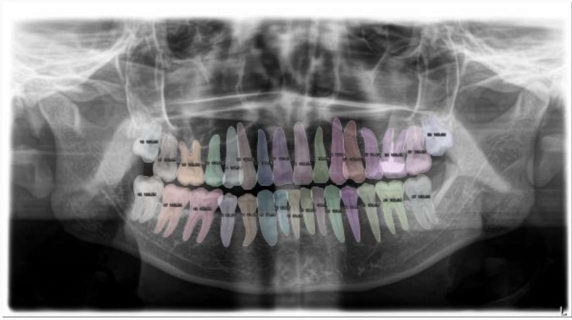
Before post-processing	After post-processing
	

Figure 11. Comparison between the raw HTC output and the post-processed result in a case with duplicate tooth predictions.

Figures 11 and 12 show that the proposed post-processing step improves the raw HTC predictions by removing duplicated masks in overlapping teeth and suppressing small fragment-like false positives. This is achieved through overlap-based duplicate suppression, local spatial consistency checking, and size-based filtering of tiny isolated masks, resulting in cleaner and more anatomically plausible final segmentation outputs.

4.3.2 Removal of small fragment-like false positives

Another recurring issue was the presence of very small mask fragments that appeared near tooth edges or between adjacent teeth. These unstable pieces usually originated from partial responses of the segmentation model and did not represent a complete tooth instance. During post-processing, such fragments were suppressed by applying minimum area constraints, connected-component cleaning, and geometric checks on mask compactness and relative tooth size. As a result, the final output became visually cleaner and less prone to spurious tooth-like regions.

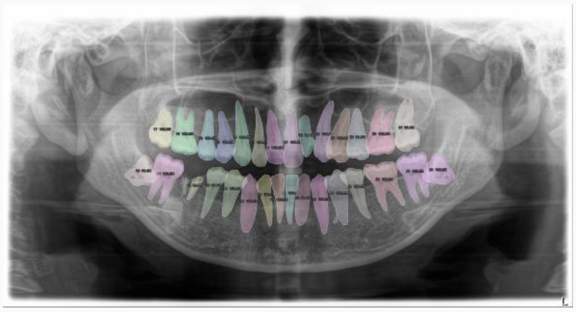
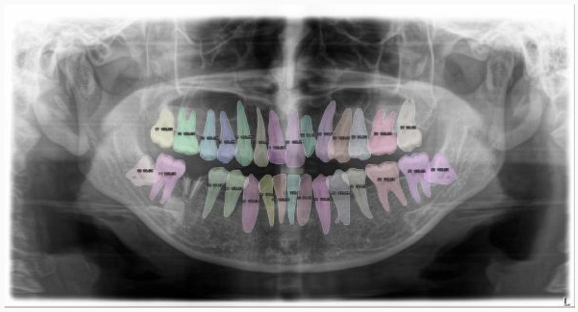
Before post-processing	After post-processing
	

Figure 12. Comparison between the raw HTC output and the post-processed result in a case with small fragment suppression.

4.4 Web Platform Demonstration : ToothInSeg

In addition to the model-level evaluation presented in Sections 4.1-4.3, the proposed pipeline was further demonstrated through the ToothInSeg web platform. Since the system architecture, API workflow, and deployment details were already described in Secti on 3.4, this section focuses only on the visual presentation of the platform and on how the final prediction results are displayed to end users.

The web interface was designed to support a straightforward clinical-style workflow in which a user uploads a panoramic radiograph, submits it for inference, and receives the final segmentation output from the best-performing model, HTC (Swin-T + FPN), together with the post-processing stage described in Section 4.3. The following figures are intended to present the main pages of the platform and the way prediction outputs are visualized within the browser.

4.4.1 Main pages of the web platform

Figure 13 should summarize the main user-facing pages of the ToothInSeg platform. A four-panel layout is recommended so that the reader can quickly understand the overall flow from image submission to case review.

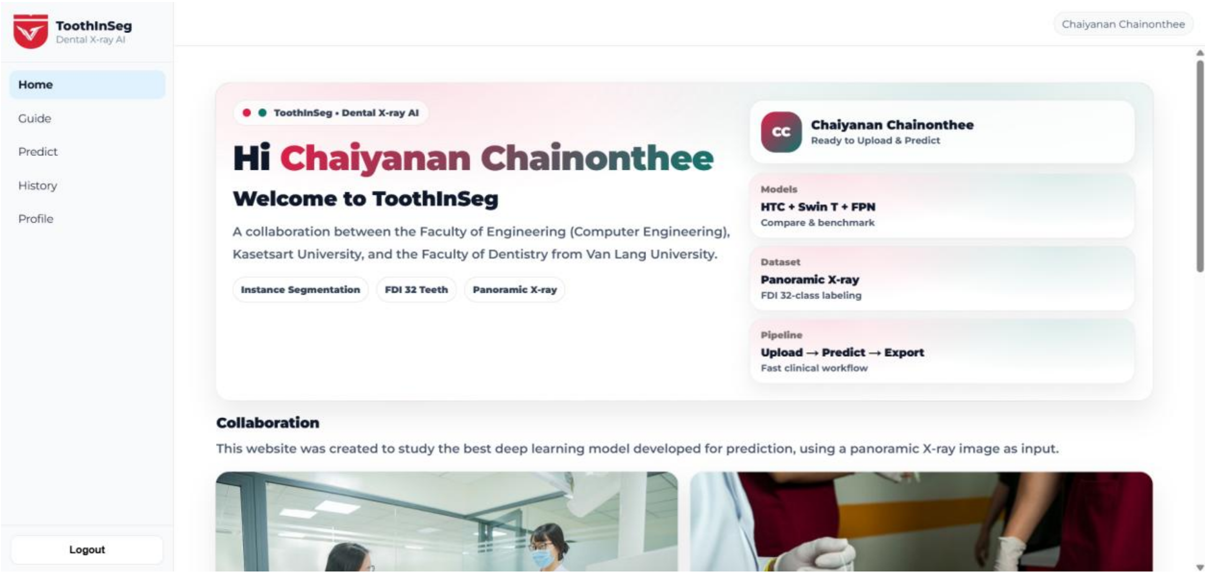


Figure 26 Home page of ToothInSeg Web platform

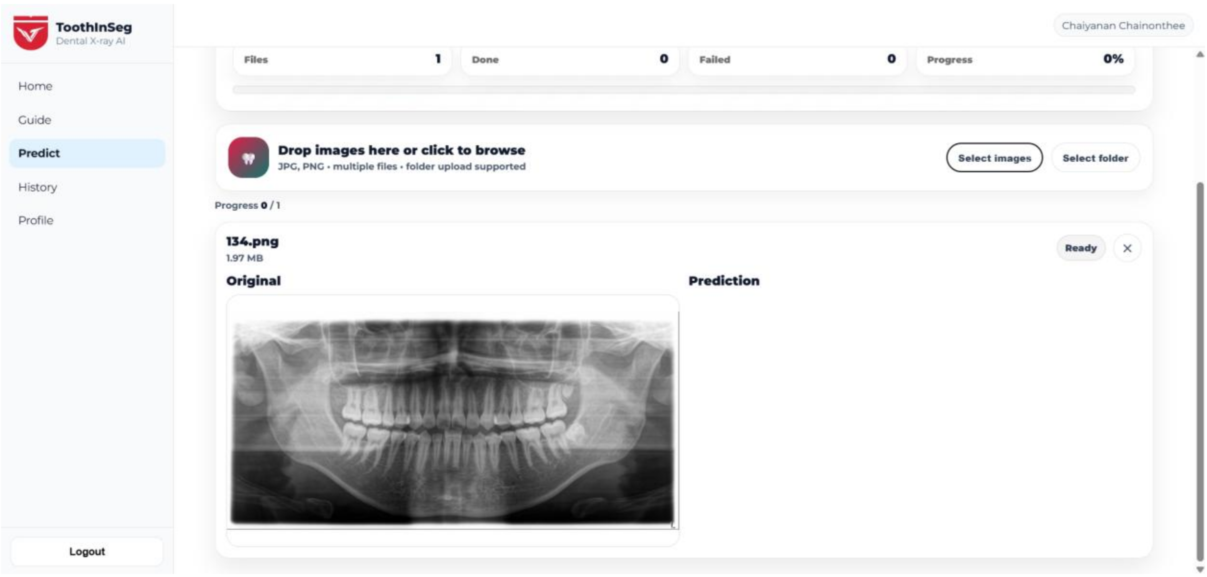


Figure 27 Image upload Page of ToothInSeg Web platform

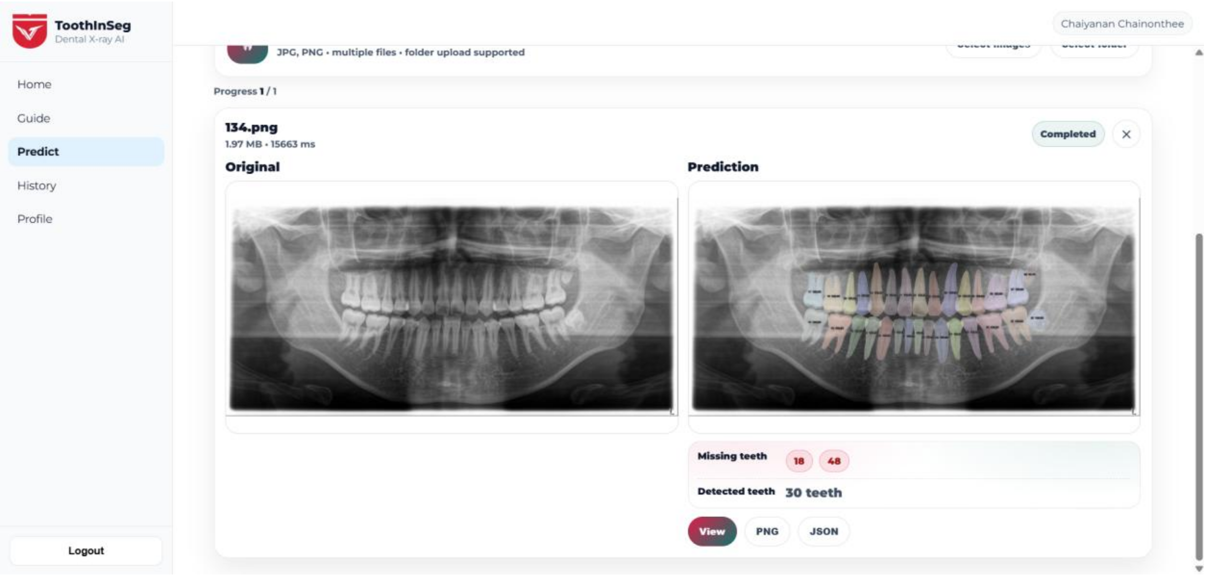


Figure 28 Prediction result page of ToothInSeg Web platform

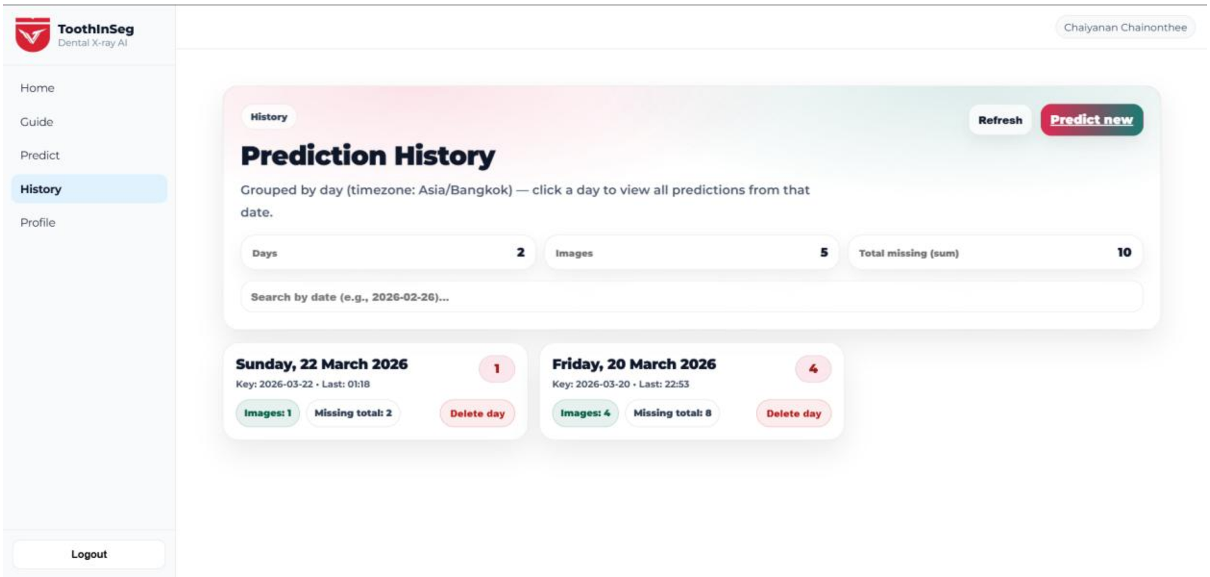


Figure 29 Prediction history page of ToothInSeg Web platform

4.4.2 Visualization of prediction outputs

Because the final goal of the platform is to make the model output easy to inspect, the most important page in this section is the prediction result interface. Figure 14 is intended to emphasize how the uploaded panoramic radiograph, the segmentation overlay, the FDI-based tooth numbering, and the case summary are presented to the user in a visually interpretable format.

Figure 30 Layout one: original panoramic image

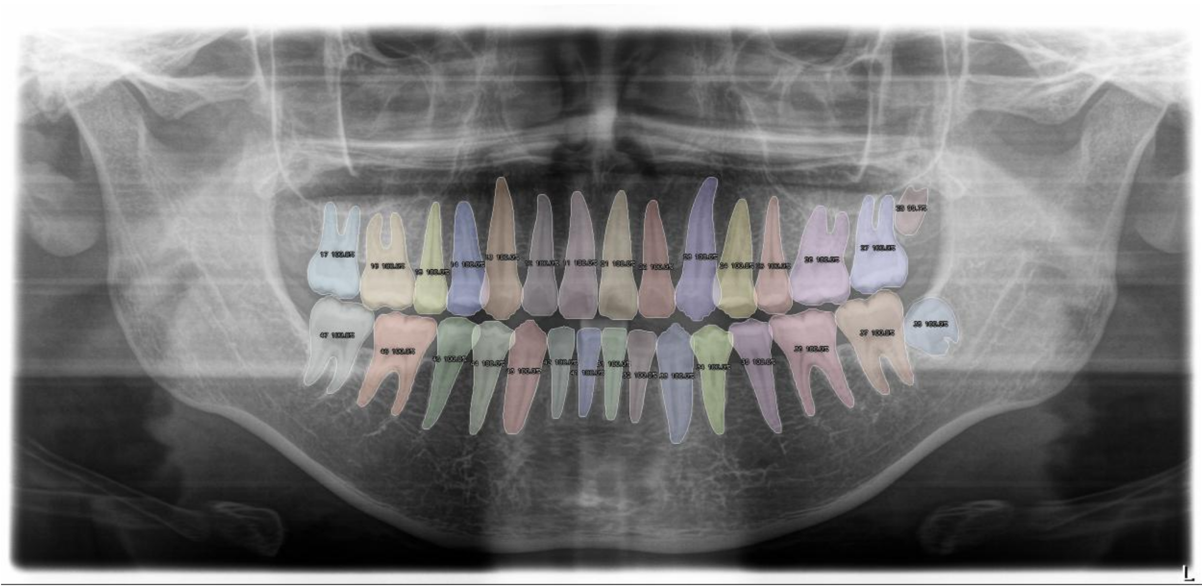


Figure 31 Lay out two: segmentation overlay

Figure 32 Zoom in or detailed result veiw

These interface figures are not intended to repeat the technical implementation details from Section 3.4. Instead, they demonstrate how the proposed segmentation pipeline is presented in a practical browser-based environment and how the final output can be reviewed by users through a simple and visually structured workflow.

Chapter 5

Conclusion

This study aimed to develop an AI-based program for analyzing 2D panoramic radiographs by performing tooth segmentation and tooth numbering. The system was designed to improve result consistency, reduce variability from manual interpretation, and support downstream digital dentistry workflows. To ensure a meaningful and interpretable comparison, the study adopted a controlled benchmarking setup using the same dataset split and evaluation protocol across methods.

In terms of organizing the dataset. This research used 1,247 panoramic images (adult patients), covering both cases with complete and missing teeth, with the dataset provided in collaboration with VanLang University. Furthermore, VanLang University provided training to the two researchers involved in this project on the fundamentals of dentistry to accurately and efficiently annotate the dataset using a locally deployed CVAT platform. and the final labels were reviewed by dental students and faculty members to improve annotation quality and consistency.

In terms of experimentation, this study benchmarked five models (4baseline model & our HTC model) on a held-out test set of 187 images (15% of the full dataset) after 80 training epochs, using the same data split, training budget, and evaluation code to ensure a fair comparison. Performance was reported using both pixel-wise metrics (Acc./Spec./Prec./Rec./F1) and COCO-style instance segmentation metrics (mAvP, AP50, AP75). Quantitatively, HTC (Swin-T + FPN) achieved the strongest overall results, obtaining F1-score = 94.96 and mAvP = 80.20, indicating a favorable balance between mask fidelity and instance-level separation on the same test set.

For qualitative assessment, three representative scenarios were presented:

1). complete dentition,

2). missing teeth,

3). crowded/overlapping teeth.

Overall, two-stage and transformer-based approaches tended to produce boundaries closer to the ground truth in regions with subtle edges. In highly crowded areas, the qualitative examples highlight the benefit of transformer-based contextual modeling for maintaining clearer separation between adjacent teeth.

Furthermore, this research added post-processing steps to the best-performing model, HTC (Swin-T + FPN), to correct errors that were still found in some cases. Focusing on two main problems:

1) duplicate masks on the same tooth.

2) small mask fragments that are not part of the actual tooth structure.

The post-processing pipeline removes unstable fragments using size and connected-component criteria with additional geometric checks and suppresses duplicates using overlap-based comparisons (overlap ratio, IoU, centroid distance, and confidence), as well as checking positional alignment along the dental arch. Before and after results show that post-processing reduces mask duplication and noise, resulting in cleaner and more anatomically plausible final results.

Finally, the end-to-end pipeline was demonstrated through the ToothInSeg web platform, which supports image upload, GPU inference, overlay visualization, and result review. The platform uses the main HTC (Swin-T + FPN) model together with the post-processing stage to present outputs that are easier to inspect in a practical workflow. Overall, the updated results and post-processing design strengthen the practicality of the proposed system for downstream digital dentistry usage.

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